

- 1 The text file `HighScore.txt` stores the players who have scored the top ten scores in a game, in descending order of score. The file stores the 3-character name of the player, and their integer score, in the order: player, score.

For example, the current top player in the text file:

FYI is the player name

10 000 is the score

The program:

- reads in the data from `HighScore.txt`
- allows the user to enter a new player name and their score
- if appropriate, inserts the new player (name and score) into the top ten
- writes the top ten players (name and score) into a new text file `NewHighScore.txt`

- (a) The program stores the players and their scores in an array of 11 elements (10 elements to be read from the file, 1 element to be inserted by the user).

Write a program to declare one or more arrays, as global data structures, to store the player names and their scores.

- (b) The procedure `ReadHighScores()` opens the file `HighScore.txt` and reads the data into the data structure(s) declared in **part 1(a)**.

Write program code to declare the procedure `ReadHighScores()`.

- (c) The procedure `OutputHighScores()` outputs all the values in the data structure(s) in the format:

PlayerName Score

For example, the first two data items: FYI 10 000
 ABC 9 092

Write program code to declare the procedure `OutputHighScores()`.

- (d) The main program should first call `ReadHighScores()` and then `OutputHighScores()`.

(i) Write the program code for the main program.

(ii) Test your program.

Take a screenshot to show the output from **part 1(d)(i)**.

- (e) The main program needs to ask the user to input a new player name and a score. If this score is in the top ten then it will create a new top ten list that includes this score.

(i) Amend the main program to ask the user to input a 3-character player name and an integer score that must be between 1 and 100 000 inclusive.

(ii) Write program code to declare a procedure that:

- takes the player name and score as parameters
- creates a new top ten list that includes the parameter if appropriate.

(iii) Amend the main program to call the procedure from **part 1(e)(ii)**.

Output the contents of the array before inserting the new player name and score, and output the contents of the array after inserting the new player name and score.

(iv) Test your program by entering the player name "JKL" and the score "9999".

Take a screenshot to show the output.

- (f) The procedure `WriteTopTen()` stores the new top ten player names and scores in a text file called `NewHighScore.txt`

Write program code to declare the procedure `WriteTopTen()`.